Mathematics in Astronomy Olympiads I

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1 Introduction

If you have recently started astronomy olympiads as a middle school student (Grades 6–8) or high school student (Grades 9–12), you will encounter several mathematical techniques and problem-solving frameworks that are not typically covered in the standard school curriculum. These concepts are essential for tackling problems in both the Junior and Senior IOAA syllabus.

This guide introduces these mathematical tools, explains their relevance in astronomy olympiads, and provides relevant practice problems to help you become familiar and comfortable with their application.

NB: This handout is designed primarily for Bangladeshi middle school students. Therefore, the presentation of concepts will closely follow the style and structure of NCTB textbooks. This ensures that new ideas are introduced in a familiar, gradual, and accessible manner, while still developing the intuition required for olympiad-level problem solving.

2 Dimension Analysis

Physical Quantities

Physics describes the world in terms of measurable properties called *physical quantities.* A physical quantity is any property that can be measured and expressed using a numerical value and a unit. Examples include length, mass, time, speed, temperature, force, and energy.

At first glance, it may seem that there are infinitely many physical quantities. However, most quantities can be expressed in terms of a small set of more basic quantities. In fact, within the SI system, all physical quantities can be built from just seven fundamental quantities.

Fundamental quantities are basic physical quantities that are independent of one another and cannot be expressed in terms of other physical quantities. They serve as the foundation from which all other (*derived*) quantities are constructed.

Table 1: Fundamental Quantities in SI Unit

Quantity	**Unit**
Length	meter
Mass	kilogram
Time	second
Electric current	ampere
Temperature	kelvin
Amount of substance	mole
Luminous intensity	candela

Many other quantities can be formed by combining fundamental quantities. These are known as *derived quan-*

tities. For example, speed is defined as distance divided by time, and force is defined through Newton's Second Law, $F = ma$. Since they are built from more basic quantities, their properties can be described in terms of the fundamental quantities.

Dimensions

Dimensions describe how a physical quantity depends on fundamental physical quantities, independent of the specific units used to measure it [1].

Table 2: Fundamental Physical Quantities and Their Dimensions

Quantity	**Dimension**
Mass	$[M]$
Length	$[L]$
Time	$[T]$

To be precise, we should distinguish dimensions and units. The dimensions of a physical quantity determine what kind of quantity it is, while a unit is a measure of a dimension. [2]
For example, somebody's height (h) may be measured in feet or meters, but both quantities share the same dimension of length; this is written as $[h] = [\text{ft}] = [\text{m}] = L$, where brackets denote dimensions.
Similarly, for speed, regardless of whether it is measured in meters per second or kilometers per hour.

$$\left[\frac{\text{distance}}{\text{time}}\right] = \frac{L}{T} = LT^{-1}$$

Understanding dimensions is useful in problem solving for several reasons. They can help predict the general form of an expression before beginning a calculation and provide a quick consistency check after obtaining a result. For example, if you're trying to derive the surface area of a sphere and find $4\pi r^3$.

$$[\text{The dimension of Area}] = L^2$$

But your answer gave L^3. You can instantly know you made a mistake.

A fundamental principle of dimensions, before getting started with problems:
Every valid physics equation must be dimensionally homogeneous, meaning all additive terms must have the same dimensions.

Problem

Check whether the relation $s = ut + \frac{1}{2}at^2$ is dimensionally correct or not, where symbols have their usual meaning.

Solution

Left-hand side (LHS):

$$[s] = L$$

Right-hand side (RHS):
First term: ut

$$[ut] = [u][t] = LT^{-1} \cdot T = L$$

Second term: $\frac{1}{2}at^2$

$$\left[\frac{1}{2}at^2\right] = [a][t^2] = LT^{-2} \cdot T^2 = L$$

Since the constant $\frac{1}{2}$ is dimensionless, it does not affect the dimensions.

Verification:

$$[s] = [ut] + \left[\frac{1}{2}at^2\right] = L$$

$$\text{LHS} = \text{RHS}$$

Conclusion:
The relation $s = ut + \frac{1}{2}at^2$ **is dimensionally correct**.

Problem

Find the dimensions of the magnetic field.

Solution

To determine the dimensions of a physical quantity such as the magnetic field (B), we start from a known relation and rearrange it in terms of the required quantity.

For example, from the Lorentz force law, $F = q(v \times B)$
Taking magnitudes, $F = qvB$
So, $B = \frac{F}{qv}$
Now substituting dimensions,

$$[B] = \frac{[F]}{[q][v]}$$

Using

$$[F] = MLT^{-2}, \quad [v] = LT^{-1}, \quad [q] = Q$$

we get

$$[B] = \frac{MLT^{-2}}{Q \cdot LT^{-1}} = MT^{-1}Q^{-1}$$

So, the dimension of magnetic field is

$$[B] = MT^{-1}Q^{-1}$$

Dimension Analysis

Dimensional analysis is based on the principle that the dimensions of a physical equation must be consistent on both sides. Since physical equations describe relationships between measurable quantities, quantities with different dimensions cannot be directly equated. This simple yet powerful idea can be used to check the validity of equations, infer relationships between variables, and sometimes even solve entire problems on its own.

So far, we have used dimensions in two ways: to check whether an equation is consistent, and to find the dimensions of an unfamiliar quantity. There is, however, another important class of problems where dimensional analysis becomes especially useful — those that ask us to find an unknown relationship or equation between physical quantities. In such problems, we usually know which quantities a result should depend on, but not the precise way in which they combine. By requiring that both sides carry the same dimensions, we can often recover the form of the relationship on our own, and sometimes solve the problem almost entirely. Let us go through one such problem to see how this works.

Problem

We will venture into the mysterious realm of quantum gravity, and use dimensional analysis to learn about black holes. Here, all of the fundamental constants will be involved. First, we can define it a little more formally, "A black hole is a region of spacetime that traps light. The boundary of this region is called the event horizon." For the purpose of this problem, the radius of the black hole means the radius around the black hole from where no light can escape.

a. Express the radius of a black hole (i.e. the light-trapping region) in terms of its mass m, gravitational constant G, and the speed of light c using only dimensional analysis.

[BDOAA 2023 T2(a)]

Worked Solution

First, as stated in the question, let us assume that the black hole radius R depends on the mass m, gravitational constant G, and speed of light c. Thus, we write

$$R \propto m^a G^b c^d$$

Now, taking dimensions on both sides, For Left Hand Side,

$$[R] = L$$

We know the dimensions of each quantity in the Right Hand Side. We know the dimensions of each quantity on the left-hand side:

$$[G] = M^{-1}L^3T^{-2}$$
$$[m] = M$$
$$[c] = LT^{-1}$$

Hence, taking dimensions, the right-hand side becomes

$$[m]^a[G]^b[c]^d = (M)^a \cdot (M^{-1}L^3T^{-2})^b \cdot (LT^{-1})^d$$
$$= M^{a-b}\, L^{3b+d}\, T^{-2b-d}$$

Equating exponents on both sides

$$a - b = 0, \quad 3b + d = 1, \quad -2b - d = 0$$

Solving the above equations, we get $a = b = 1$ and $d = -2$, from which we understand that

$$R \sim \frac{Gm}{c^2} \;\; \Rightarrow \;\; R_s = \frac{2Gm}{c^2}$$

Now that we have seen how to construct a relationship between physical quantities using only their dimensions, let us approach a similar problem from IOAA 2022.

Problem: IOAA 2022 [T-1]

In Max Planck's system of natural units of measurements, commonly used in cosmology, all units are expressed in terms of 4 fundamental constants: speed of light (c), universal gravitational constant (G), reduced Planck constant $\left(\hbar = \frac{h}{2\pi}\right)$ and Boltzmann constant (k_B). Also, ($4\pi\varepsilon_0$) is included in this list of constants.

Using dimensional analysis, find expressions for

- Planck length (L_p),
- Planck time (t_p),
- Planck mass (m_p),

- Planck temperature (T_p) and
- the Planck charge (q_p).

Solution

First, write the dimensions of the fundamental constants in terms of length (L), mass (M), time (T), temperature (Θ) and charge (Q).

$$c = [L^1 M^0 T^{-1}]$$

$$G = [L^3 M^{-1} T^{-2}]$$

$$\hbar = [L^2 M^1 T^{-1}]$$

$$k_B = [L^2 M^1 T^{-2} \Theta^{-1}]$$

Planck Length

To obtain a quantity with the dimensions of length, we seek a combination of G, $\hbar$, and c whose dimensions reduce to L^2.

$$G\hbar = [L^5 M^0 T^{-3}]$$

Dividing by c^3 removes the remaining powers of time:

$$\frac{G\hbar}{c^3} = [L^2 M^0 T^0]$$

Taking the square root gives a quantity with dimensions of length.

$$\therefore \quad L_p = \sqrt{\frac{G\hbar}{c^3}}$$

Planck Mass

Next, we look for a combination with dimensions of M^2.

$$\frac{\hbar c}{G} = [L^0 M^2 T^0]$$

Taking the square root gives the Planck mass.

$$\therefore \quad m_p = \sqrt{\frac{\hbar c}{G}}$$

Planck Time

Time can be obtained by dividing a length scale by a velocity scale. Since c has dimensions of velocity, we use

$$\frac{L_p}{c} = [L^0 M^0 T^1].$$

Substituting the expression for L_p,

$$t_p = \frac{1}{c}\sqrt{\frac{G\hbar}{c^3}} = \sqrt{\frac{G\hbar}{c^5}}.$$

Hence,

$$\therefore \quad t_p = \sqrt{\frac{\hbar G}{c^5}}$$

Planck Temperature

The Boltzmann constant relates energy to temperature:

$$k_B T \sim \text{energy}.$$

Therefore, the natural temperature scale is obtained by dividing the Planck energy $m_p c^2$ by k_B:

$$T_p = \frac{m_p c^2}{k_B}.$$

Substituting the expression for m_p,

$$T_p = \frac{c^2}{k_B}\sqrt{\frac{\hbar c}{G}} = \sqrt{\frac{\hbar c^5}{G k_B^2}}.$$

Thus,

$$\therefore \quad T_p = \sqrt{\frac{\hbar c^5}{G k_B^2}}$$

Planck Charge

For charge, we first note the dimensions of the electric constant:

$$4\pi\varepsilon_0 = [L^{-3} M^{-1} T^2 Q^2].$$

Also,

$$\hbar c = [L^3 M^1 T^{-2}].$$

Multiplying these quantities leaves only Q^2:

$$(4\pi\varepsilon_0)(\hbar c) = [Q^2].$$

Taking the square root gives the Planck charge:

$$\therefore \quad q_p = \sqrt{4\pi\varepsilon_0\, \hbar c}.$$

Now that we have worked through two examples together, the following problems are left for you to try on your own. Apply the same process as before, and check your scaling carefully.

IOAA 2018, Theory T11
IOAA 2025, Theory T03

Now let us solve another type of problem using dimension analysis.

Problem

A circle of rope is spinning in outer space with an angular velocity ω_0. Transverse waves on the rope have speed v_0, as measured in a rotating reference frame where the rope is at rest. If the angular velocity of the rope is doubled, what is the new speed of transverse waves?

Solution

To solve this problem using dimensional analysis, we consider which physical quantities could possibly affect the speed of transverse waves. The result could depend on the rope's length L, mass per unit length λ, and angular velocity ω_0. It could also depend on the tension in the rope, but since the tension is determined by the balance of centrifugal effects, it is not an independent variable.
Thus, the relevant quantities are

$$[L] = \mathrm{m}, \quad [\lambda] = \mathrm{kg\,m^{-1}}, \quad [\omega_0] = \mathrm{s^{-1}}.$$

Since λ is the only quantity involving mass, it cannot contribute to the final expression for a speed, because there is no other independent quantity available to cancel the mass dimension.
Therefore, the only possible combination with dimensions of speed is

$$v_0 \sim L\omega_0.$$

Here, the symbol $\sim$ indicates equality up to a dimensionless constant, which cannot be determined using dimensional analysis alone. In practice, this constant is usually not very large or very small, so $L\omega_0$ provides a reasonable estimate of v_0.
However, even when the exact constant is unknown, dimensional analysis correctly captures the scaling behaviour. In particular, since $v_0 \sim L\omega_0$, if the angular velocity ω_0 is doubled, the wave speed also doubles, giving the new speed $2v_0$.

[Kevin Zhou, *Physics Olympiad Handouts: Problem Solving I: Mathematical Techniques*]

3 Unit Conversion

3.1 What Are Units?

Every physical quantity is measured by comparing it against an agreed upon reference. A **unit** is simply the name we give to this reference for a particular quantity — for example, we measure length in *meters* (m), mass in *kilograms* (kg), and time in *seconds* (s).

Behind every unit lies a **standard**: a fixed, agreed-upon amount corresponding to exactly 1.0 of that unit. Saying a table is 2 meters long means it is twice the length of the standard meter. We are free to define a standard however we like, it only needs to be sensible, practical, and reproducible, so that scientists everywhere can agree on it. Once a standard is fixed, we still need ways to measure quantities far too large or too small to compare with it directly — we cannot lay a meter stick against an atom, nor stretch one to a star. For these we rely on **indirect measurements** that ultimately trace back to the same standard.

This is why a single quantity can be expressed in many different units — a distance might be given in meters, kilometers, or even light-years — while still describing the same physical thing.

3.2 Units vs. Dimensions

At this point, it is worth returning to a distinction we touched upon earlier and making it precise, since the two ideas are easy to confuse.
A **dimension** tells us *what kind* of physical quantity we are dealing with, whether it is a length, a mass, a time, or some combination of these. A **unit**, on the other hand, tells us *how much* of that quantity we have, measured against a particular chosen standard. In short: dimension describes the nature of a quantity, while unit describes its scale.

A single dimension can be measured using many different units. Length, for instance, has the dimension $[L]$ regardless of whether we choose to measure it in meters, feet, light-years, or parsecs:

$$[\mathrm{m}] = [\mathrm{ft}] = [\mathrm{ly}] = [\mathrm{pc}] = L$$

Each of these units corresponds to a different *amount* of length, but all of them describe the same underlying dimension.
This distinction has an important consequence: converting between units — say, from meters to light-years — never changes the dimension of a quantity. It only rescales the numerical value used to express it. Dimensional analysis, as we saw in Section 2, is therefore unaffected by which units we choose to work in; it only cares about the dimension itself.

3.3 Systems of Units

Now that we understand what units are, let us look at how they are organized into complete systems.
In 1971, the 14th General Conference on Weights and Measures adopted seven physical quantities as **base quantities**, forming the foundation of the **International System of Units**, abbreviated **SI** (from its French name, *Système International*) and commonly known as the metric system. Every other unit in the SI system, called a **derived unit**, is built from these seven base units.

Base Quantity	**Unit Name**	**Symbol**
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Table 3: The seven SI base units

As an example of a derived unit, consider power, whose SI unit is the **watt** (W). Since power is energy delivered per unit time, and energy itself is built from mass, length, and time, the watt can be written entirely in terms of base units:

$$1\ \mathrm{W} = 1\ \mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-3}$$

This is a useful habit to develop: any derived unit can always be broken down into a combination of the seven base units, and doing so is often the fastest way to check or convert it.

Prefixes for SI Units

Since physical quantities can range from the extremely small to the extremely large, SI units are often combined with standard **prefixes** that scale them by powers of ten. For example, a kilometer (km) is 10^3 meters, while a nanometer (nm) is 10^{-9} meters. The adjacent table lists the most common SI prefixes. These prefixes will appear often when converting between units, particularly in Section 3.4, so it is worth becoming comfortable with the most common ones — especially kilo-, centi-, milli-, micro-, and nano-.

Factor	**Prefix**	**Symbol**
10^{18}	exa-	E
10^{15}	peta-	P
10^{12}	tera-	T
$\mathbf{10^{9}}$	**giga-**	**G**
$\mathbf{10^{6}}$	**mega-**	**M**
$\mathbf{10^{3}}$	**kilo-**	**k**
10^{2}	hecto-	h
10^{1}	deka-	da
10^{-1}	deci-	d
$\mathbf{10^{-2}}$	**centi-**	**c**
$\mathbf{10^{-3}}$	**milli-**	**m**
$\mathbf{10^{-6}}$	**micro-**	$\boldsymbol{\mu}$
$\mathbf{10^{-9}}$	**nano-**	**n**
$\mathbf{10^{-12}}$	**pico-**	**p**
10^{-15}	femto-	f
10^{-18}	atto-	a

Common SI prefixes

Units in Astronomy

While SI units are standard throughout physics, astronomy frequently works with distances, masses, and luminosities so extreme that expressing them in meters, kilograms, or watts becomes impractical. For this reason, astronomers commonly use a set of unit systems and reference quantities better suited to astronomical scales.
These are not part of the SI system, but they are dimensionally consistent with it — an astronomical unit is still a length, and

Quantity	Unit	Approximate SI Value
Distance	Astronomical Unit (AU)	1.496×10^{11} m
Distance	Light-year (ly)	9.461×10^{15} m
Distance	Parsec (pc)	3.086×10^{16} m
Mass	Solar mass ($M_\odot$)	1.989×10^{30} kg
Radius	Solar radius ($R_\odot$)	6.96×10^{8} m
Luminosity	Solar luminosity ($L_\odot$)	3.83×10^{26} W

Table 4: Common reference units in astronomy

a solar mass is still a mass. They are simply chosen because their scale matches the objects astronomers usually study, such as planetary orbits or stellar masses, making numbers easier to work with.

You may also encounter the **CGS system** (centimeter–gram–second) in older astronomical literature and some areas of astrophysics, such as stellar structure and radiative transfer, where quantities like density are often quoted in $\mathrm{g\,cm^{-3}}$ rather than $\mathrm{kg\,m^{-3}}$. Regardless of which system a problem uses, the underlying physical dimensions remain the same, only the numerical values and unit names change.

3.4 Changing Units

We often need to express a physical quantity in a different unit than the one it was originally given in. The standard method for doing this is called **chain-link conversion**. The idea is simple: we multiply the original quantity by a **conversion factor**, which is a ratio of two equivalent units that is numerically equal to 1.
For example, since 1 minute and 60 seconds represent the same interval of time, we can write

$$\frac{1 \text{ min}}{60 \text{ s}} = 1 \qquad \text{and} \qquad \frac{60 \text{ s}}{1 \text{ min}} = 1.$$

Either ratio can serve as a conversion factor. Note that this is different from simply writing $\frac{1}{60} = 1$ or $60 = 1$; the units are just as important as the numbers, and must always be carried along together.
Since multiplying any quantity by 1 leaves it unchanged, we are free to introduce a conversion factor wherever it helps us. The trick is to choose the factor so that the unwanted unit cancels out, leaving only the unit we want. For example, to convert 2 minutes into seconds:

$$2 \text{ min} = (2 \text{ min})(1) = (2 \cancel{\text{min}}) \left(\frac{60 \text{ s}}{1 \cancel{\text{min}}} \right) = 120 \text{ s}.$$

If a conversion factor is set up so that the unwanted unit does *not* cancel, the factor has simply been inverted, so flipping it and trying again will fix this. In every case, units behave exactly like algebraic quantities: they can be multiplied, divided, and cancelled just as numbers or variables can.

Problem

The average distance between the Earth and the Sun is 1 astronomical unit (AU), where 1 AU = 1.496×10^8 km. Express this distance in meters.

Worked Solution

We are given a distance in astronomical units and asked to express it in meters. Since 1 AU = 1.496×10^8 km and 1 km = 10^3 m, we can chain together two conversion factors, choosing each one so that the unwanted unit cancels:

$$1 \text{ AU} = (1 \cancel{\text{AU}}) \left(\frac{1.496 \times 10^8 \cancel{\text{km}}}{1 \cancel{\text{AU}}} \right) \left(\frac{10^3 \text{ m}}{1 \cancel{\text{km}}} \right) = 1.496 \times 10^{11} \text{ m}.$$

Conclusion: 1 AU = 1.496×10^{11} m.

4 Recommended Video

Dimensional Analysis: Video Lecture

A walkthrough of a problem covered in this handout. Watching this will help you better understand how to approach problems through dimension analysis.

▶ **Watch on YouTube**

5 Further Reading

For readers who wish to explore these topics in greater depth, the following resources are recommended:

- Fahim Rajit Hossain, জ্যোতির্বিজ্ঞানের যতকিছু, অলিম্পিয়াড ও অন্যান্য (2025) — A2: Dimension Analysis.
- Kevin Zhou Physics Olympiad Handouts: Problem Solving I: Mathematical Techniques

References

[1] National Curriculum and Textbook Board (NCTB), *Physics: Classes IX–X*, 2023

[2] Kevin Zhou, *Physics Olympiad Handouts: Problem Solving I: Mathematical Techniques*,